# When Topology Decides the Load:
## A Structural Attractor in Graph Routing, Found Through Classical-Physics Heuristics

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**Abstract**

Load balancing on a graph is usually framed as an algorithm-design problem: a better router yields a better load distribution. We report evidence that, over a wide range of topologies, the load distribution is instead largely fixed by the structure of the graph and the traffic demand, leaving any reasonable congestion-aware router little freedom—and we identify a topological metric that predicts how much freedom remains. We reached this through an unusual route. Treating a packet as a particle under a classical force, we built three independent routing cores—a Newtonian loss-reaction, an Archimedean buoyancy, and a Pascalian pressure diffusion—sharing a search engine but no mechanism. The three produce near-identical per-edge load distributions (cosine 0.99), significantly tighter than blind controls such as ECMP (0.87), and reach statistical equivalence (TOST) with the engineered balancers CONGA and DRILL on most of fifteen topologies. A sensitivity analysis is candid about the limits of that equivalence: at a strict margin (0.5 percentage points of drop rate) only a few topologies remain equivalent, and they are precisely the ones where the structural attractor is strongest. We argue, via a cut-based structural argument, that this is no coincidence: the demand fixes the flow across every cut, the topology's cut structure pins most of the load distribution, and any congestion-aware rule—physical or engineered—shapes the small residual identically. A purely structural ratio, tube/sp (room to manoeuvre among near-shortest paths, computable before deployment), predicts where the residual is large enough for routers to differ; the correlation is robust to leave-one-out and strengthens when the loosest-fitting topologies are removed ($r$ rising from 0.78 to 0.89). The physics, in the end, was a discovery instrument rather than the result: three classical analogies converged because the problem, not the analogy, drove them to the same place. We report our negative results, an audit retracting an earlier inflated advantage, and the margin sensitivity in full.

## 1 Introduction

Network load balancing is a mature engineering discipline. Production systems such as CONGA [1] and DRILL [2] spread traffic across a network using carefully designed mechanisms: per-flowlet rerouting, multi-path candidate enumeration, local queue sampling, and centrally or distributively maintained congestion state. These systems work well, and improving on them is hard.

This paper does not try to improve on them. It asks a different and, we think, more interesting question:

> *How much of what engineered load balancers achieve can be recovered from classical physics alone—from a packet treated as a particle obeying a couple of elementary mechanical laws?*

The motivation is partly methodological. An expert in routing reasons within the conceptual vocabulary of routing; a physicist handed the same problem reasons about gradients, forces, and conservation laws. Occasionally the second framing reaches a familiar destination by an unfamiliar and more economical road, and the comparison is informative regardless of which arrives first. Our interest is therefore not a horse race but a question of *parsimony*: how little mechanism suffices to reach the operating point that elaborate engineering reaches.

### 1.1 The model, in one paragraph

We treat a packet as a particle descending a potential field defined on the network graph. The particle feels two forces. The first is a *gradient*: it rolls toward lower latency and lower congestion, subject to a hop budget that lets it take a modestly longer, emptier detour rather than forcing every packet onto the single shortest path. The second is a *reaction* in the sense of Newton's third law: when a packet is dropped on a congested edge, it pushes back, depositing a decaying "scar" that steers subsequent packets away from edges that have just failed. That is the entire router—two terms, no central controller, no per-flow state, no precomputed path tables, sixty lines of code.

### 1.2 What we find

Our central result is deliberately stated as an *equivalence*, not a victory:

1. **Two terms suffice.** The two-term core is the survivor of an ablation: an earlier model carried eight physically-motivated layers (effective mass, a thermostat, a temporal ripple, buoyancy), and measurement showed that only the gradient and the Newton-III scar earn their keep—the rest are inert or mildly harmful (Section 4).

2. **It matches engineered routers.** Tested as a statistical equivalence (TOST, margin 2 percentage points) across fifteen topologies in a flow-level simulator, the core is equivalent to DRILL on twelve of fifteen, and equivalent to CONGA once CONGA is given a fair path budget. It reaches the same operating point CONGA reaches only after enumerating and scoring sixteen to thirty-two candidate paths per flow (Sections 5–6).

3. **A topological metric predicts when.** A quantity computable from the graph alone, the tube/sp ratio (room to manoeuvre among near-shortest paths), predicts where the equivalence holds and where engineered multi-path routing keeps a genuine edge (Pearson $r = 0.78$); the two real WAN backbones, with low tube/sp, are exactly the predicted exceptions (Section 9).

The upshot is that two classical forces, with none of the apparatus of modern load balancers, reach the same place those balancers reach, on the topologies where the geometry leaves room to do so—and that the qualifying clause is itself predictable.

### 1.3 A note on method

This paper reports its negative results and its own corrections in full. The eight-layer model is presented and then dismantled; an early, larger advantage over CONGA is shown to have been an artifact of an unfair path budget and is retracted (Section 6). We regard this not as throat-clearing but as the substance of the contribution: a claim that physics matches engineering is worth precisely as much as the adversarial scrutiny it has survived, and we would rather report an honest equivalence than an inflated win. Every headline figure here has been checked against the strongest version of its baseline we could build.

## 2 Related Work

**Engineered load balancing.** Modern multipath load balancers spread traffic to avoid hot spots. ECMP [3] splits flows across equal-cost shortest paths by hashing, with no congestion awareness. CONGA [1] adds that awareness: it enumerates a set of candidate paths and selects among them using in-network congestion feedback, rerouting at flowlet granularity. DRILL [2] pushes the decision to the switch and the microsecond, making per-packet local choices based on queue occupancy with minimal state. These systems define the operating points we measure against. Our aim is not to outperform them—on the topologies where they are strong, they remain strong—but to ask how much of their behaviour is recoverable from a far simpler, physically-motivated rule.

**Physics- and nature-inspired routing.** Using physical or biological analogies for routing has a long history: ant-colony optimisation, electrical-network and resistance analogies, potential- and field-based forwarding, and physarum-inspired path formation. These approaches typically introduce a mechanism and demonstrate that it works. Our contribution differs in emphasis in two ways. First, we run the analogy in reverse as an ablation: rather than showing that added physics helps, we add a great deal of physics and then measure how much can be removed, arriving at a two-term residue. Second, we frame the outcome as a statistical equivalence against strong engineered baselines, rather than as a demonstration that the physical method is novel or superior.

**Equivalence testing.** Most networking evaluations test for a difference (one method beats another). We instead test for *equivalence*, using TOST (two one-sided tests), which asks whether two methods are indistinguishable within a stated margin. This is the statistically appropriate tool when the claim is "as good as", not "better than", and it guards against the familiar error of reading a non-significant difference as evidence of equivalence. To our knowledge, equivalence testing is uncommon in the routing-evaluation literature, and its use here is deliberate: the entire thesis is one of equivalence with parsimony.

**Applicability conditions.** A recurring weakness of method papers is silence about when the method fails. We attach to our result an explicit, topology-only predictor (the tube/sp ratio, Section 9) that says in advance where the equivalence holds and where engineered routing retains an edge. This places the contribution closer to the spirit of complex-systems and network-science work, where structural predictors of dynamical behaviour are themselves the object of study, than to a conventional systems-performance paper.

# 3 The Two-Term Physical Core

We model a packet as a classical particle descending a potential field defined on the network graph. The particle is subject to two forces, and only two. We state them first, then justify by ablation (Section 4) that no further mechanism is needed.

## 3.1 The potential

Each directed traversal of an edge $(u, v)$ carries a cost

$$\phi(u,v) \;=\; \alpha\,\ell(u,v) \;+\; \beta\,\frac{x(u,v)}{c(u,v)} \;+\; \gamma\,s(u,v), \tag{1}$$

where $\ell$ is edge latency, $x/c$ is the instantaneous load-to-capacity ratio (congestion), and $s$ is a persistent *scar* term defined below. The three weights $(\alpha, \beta, \gamma)$ are the only tunable quantities of the model. All three are local: computing $\phi$ needs no global state beyond the edge itself.

## 3.2 Force 1: gradient descent

The particle follows the downhill direction of $\phi$ from source to destination, subject to a hop budget: it may use a path no longer than $\lceil \alpha_{\text{budget}}\, h_{\text{sp}} \rceil$ hops, where $h_{\text{sp}}$ is the shortest-path hop count. The minimum-potential path under this budget is found exactly by a dynamic program over (hops used, node), in $O(k\,|E|)$ time with $k = \lceil \alpha_{\text{budget}} h_{\text{sp}} \rceil$. The budget is what lets the particle take a slightly longer, less congested detour instead of forcing every packet onto the single shortest path—the mechanism by which load spreads.

## 3.3 Force 2: Newton's third law (the scar)

The gradient term reacts to congestion that is *present*. It cannot react to congestion that has just *caused a loss*: once a packet is dropped, the offending edge's instantaneous load may already have fallen. We

**Algorithm 1:** EMMET-core routing step (two-term physical router)

**Input:** graph $G = (V, E)$ with per-edge latency $\ell$, load $x$, capacity $c$; scar field $s$; source $a$, destination $z$; weights $\alpha, \beta, \gamma$; budget factor $\alpha_b$
**Output:** path $a \rightsquigarrow z$
$h_{\mathrm{sp}} \leftarrow$ shortest-path hop count $a \rightarrow z$;
$k \leftarrow \lceil \alpha_b\, h_{\mathrm{sp}} \rceil$; `// hop budget`
**for** *each edge* $(u, v)$ **do**
  $\phi(u,v) \leftarrow \alpha\, \ell(u,v) + \beta\, x(u,v)/c(u,v) + \gamma\, s(u,v)$;
**end**
$P \leftarrow$ min-$\sum \phi$ path of length $\leq k$ via DP over (hops, node); `//` $O(k|E|)$
**return** $P$;

`// On a drop at edge` $(u, v)$ `while forwarding:`
$s(u,v) \leftarrow s(u,v) + 1$; `// Newton-III reaction`
`// Once per step, everywhere:`
$s \leftarrow \rho\, s, \quad \rho = 2^{-1/T_{1/2}}$; `// scars fade`

close this gap with a reaction term inspired by Newton's third law—to every action a packet undergoes, the network responds with an equal and opposite reaction on the packets that follow.

Concretely, when a packet is dropped on edge $(u, v)$, that edge's scar is incremented, $s(u, v) \leftarrow s(u, v) + 1$. The scar enters the potential (1) with weight $\gamma$, so subsequent particles feel a repulsive reaction steering them away from edges that recently failed—information the instantaneous congestion term does not carry.

Scars are not permanent: at each step they decay, $s(u, v) \leftarrow \rho\, s(u, v)$ with $\rho = 2^{-1/T_{1/2}}$ and half-life $T_{1/2}$. A corridor that stops failing reopens on its own, and at rest (no losses) the scar field vanishes and the router reduces to pure gradient descent. The mechanism is self-regulating: it acts only when and where the network is actually breaking. This is the only memory in the model, and—as the ablation will show—the only added mechanism that earns its keep.

### 3.4 What is deliberately absent

This two-term core is the survivor of a larger model. Earlier versions of EMMET endowed the packet-particle with extra physical attributes: an effective mass that grew with traversed congestion, a network thermostat modulating the congestion weight, a temporal ripple diffusing loss memory over several steps, and an Archimedes-style buoyancy term. Each was physically motivated and individually plausible.

A systematic ablation (Section 4) removed them one at a time and measured the cost. The result was unambiguous: the gradient and the Newton-III scar account for essentially all of the performance, while the remaining mechanisms ranged from negligible to mildly harmful—removing the ripple and the thermostat slightly *improved* results. We therefore report the two-term core as the model and treat the richer apparatus as a cautionary tale about physically-seductive terms that do not survive measurement. The reference implementation is 60 lines of Python.

## 4 Ablation: Why Two Terms

Section 3 asserted that only two mechanisms survive scrutiny. This section is the evidence. EMMET did not begin parsimonious; it began as an exercise in taking the particle metaphor seriously, and accreted eight physical layers. We removed them one at a time and measured what each was worth.

### 4.1 The candidate layers

Beyond the gradient (which is the irreducible substrate—without it there is no router) the full model carried four added mechanisms:

Table 1: Change in total losses when each layer is removed from the full model (positive = the layer was helping; negative = it was hurting). GÉANT, trunk failure, 30 seeds. The Newton-III scar is the only layer whose removal clearly hurts; the ripple's removal slightly *helps*.

| Layer removed | moderate load | saturated load |
|---|---|---|
| Temporal ripple | −0.4% | −0.5% |
| Thermostat | 0.0% | 0.0% |
| Effective mass | +4.2% | +1.6% |
| Newton-III scar | +16.7% | +6.4% |
| All four (bare core) | +38.7% | +9.8% |

- **Newton-III scar** (loss-reaction memory, Section 3).
- **Effective mass**: a packet's influence on the potential grew with the congestion it had already traversed, modelling inertia.
- **Thermostat**: a global network "temperature" that scaled the congestion weight $\beta$ as mean utilisation rose.
- **Temporal ripple**: loss memory deposited not as a single increment but diffused over several subsequent steps, like a wave.

### 4.2 Protocol

We evaluated the full model and four leave-one-out variants (each removing a single layer), plus a bare core (gradient only, all four layers removed), against the trivial-but-stateless DRILL baseline on the GÉANT backbone under trunk-link failure. Two load regimes were used—moderate (capacity drawn from $[2, 4]$) and saturated ($[2, 2]$)—across the failed-link ensemble with 30 seeds each. The quantity of interest is the *change in total losses when a layer is removed* from the full model: a positive change means the layer was doing useful work; a negative or zero change means it was inert or harmful.

### 4.3 Result: only the scar earns its keep

Table 1 reports the change in losses on removal of each layer, in both regimes.

The reading is blunt:

- The **Newton-III scar** is the one added layer that earns its keep: removing it raises losses by 6.4–16.7%. It is doing the real work.
- **Effective mass** contributes marginally (1.6–4.2%).
- The **thermostat** has *no measurable effect* (0.0% in both regimes): it is pure decoration.
- The **temporal ripple** is mildly *harmful*: removing it slightly improves results. The most physically evocative layer—a wave of loss memory rippling outward—was a net negative.

Removing all four at once (the bare gradient core) costs 9.8–38.7%, confirming that the gradient alone is not enough: it needs the scar. But it needs *only* the scar. The mass, thermostat, and ripple together account for almost nothing, and the ripple actively hurts.

### 4.4 Interpretation

This is the parsimony result that gives the paper its title. Of the eight-layer apparatus, the data retain exactly two mechanisms—gradient descent and the Newton-III scar—and discard the rest. Crucially, the discarded layers were not obviously bad ideas: effective mass, thermal scaling, and wave-like diffusion are all physically natural, and each had a plausible story. They simply did not survive measurement. We take this as a methodological lesson worth stating explicitly: in physics-inspired modelling, the seductiveness of a mechanism is no evidence of its usefulness, and only ablation tells them apart. The two-term core of Section 3 is what remained standing.

## 5 Equivalence with Engineered Routers

The central claim of this paper is deliberately modest and precisely testable:

> *The two-term physical core reaches the operating point of purpose-built load balancers across most topologies, using a fraction of their machinery; and a topological metric predicts the cases where it does not.*

We test this as a statistical *equivalence* claim, not a superiority claim. The distinction matters: showing that two methods are indistinguishable requires a different test than showing one beats the other, and conflating the two is a common error.

### 5.1 Design

We compare the two-term core (gradient + Newton-III, no mass) against two engineered baselines: **CONGA**, a multi-path congestion-aware balancer that scores $K$ shortest paths and selects the least congested, and **DRILL**, a near-stateless balancer that makes per-hop local decisions. ECMP and shortest-path serve as weak controls the physical core is expected to beat.

All routers are evaluated in a flow-level simulator: a flow is born, lives for several ticks injecting sustained load along its route, then dies. This matters because routing behaviour differs between the packet-bullet regime (where DRILL's per-hop reactivity shines) and the flow regime (where choosing a whole path well at birth matters more). We report the flow regime as the more realistic of the two for backbone traffic.

We sweep **15 topologies** spanning four families: regular lattices (grids $5 \times 5$ to $12 \times 12$), small-world (Watts–Strogatz, several sizes and degrees), scale-free (Barabási–Albert), and two real WAN backbones (GÉANT, Abilene). Each topology is run with $n = 20$ seeds. The performance metric is the drop rate (fraction of flow-ticks that hit an overloaded edge; lower is better).

### 5.2 Equivalence test

For each topology we test equivalence between the core and each baseline with **TOST** (two one-sided tests) on the paired per-seed drop rates, with an equivalence margin $\delta = 2$ percentage points. TOST declares equivalence only when the 90% confidence interval of the mean difference lies entirely within $[-\delta, +\delta]$: that is, when we can reject *both* the hypothesis that the core is worse by more than $\delta$ *and* that it is better by more than $\delta$. This is a stricter, more honest bar than observing overlapping confidence intervals.

### 5.3 Results

Table 2 reports the per-topology drop rates and TOST verdicts. The core is statistically equivalent to DRILL on **12 of 15 topologies**. Against CONGA at its default budget ($K = 4$) the formal equivalence count is lower (5 of 15), but—crucially—most of the non-equivalences are cases where the core is *better*, not worse: on the larger grids the core's drop rate is 0.002–0.005 versus CONGA's 0.034–0.066. We examine this apparent advantage critically in Section 6, where it turns out to be an artifact of CONGA's path budget rather than a genuine physical superiority.

The scale-free topologies (Barabási–Albert) sit near zero drop for every router: their hub structure leaves little congestion to redistribute, so equivalence there is trivial and we do not lean on it. The informative cases are the grids, the small-world graphs, and the real backbones.

Table 2: Drop rate (lower is better) of the two-term core versus engineered baselines across 15 topologies, flow regime, $n = 20$ seeds. "≡D" and "≡C" mark TOST equivalence ($\delta = 2$pp) with DRILL and $\text{CONGA}_{K=4}$ respectively.

| Topology | core | $\text{CONGA}_{K=4}$ | DRILL | ≡D | ≡C |
|---|---|---|---|---|---|
| Grid 5 | 0.018 | 0.029 | 0.032 | | |
| Grid 6 | 0.011 | 0.024 | 0.023 | yes | yes |
| Grid 7 | 0.011 | 0.027 | 0.022 | yes | |
| Grid 8 | 0.005 | 0.034 | 0.015 | yes | |
| Grid 10 | 0.003 | 0.050 | 0.019 | | |
| Grid 12 | 0.002 | 0.066 | 0.009 | yes | |
| WS $n30\,k4$ | 0.016 | 0.033 | 0.004 | yes | |
| WS $n50\,k4$ | 0.023 | 0.040 | 0.009 | | |
| WS $n50\,k6$ | 0.005 | 0.007 | 0.000 | yes | yes |
| WS $n80\,k4$ | 0.022 | 0.037 | 0.008 | yes | |
| BA $n50\,m2$ | 0.003 | 0.001 | 0.000 | yes | yes |
| BA $n50\,m3$ | 0.000 | 0.000 | 0.000 | yes | yes |
| BA $n80\,m2$ | 0.001 | 0.001 | 0.000 | yes | yes |
| GÉANT | 0.057 | 0.042 | 0.053 | yes | |
| Abilene | 0.057 | 0.042 | 0.053 | yes | |

Table 3: Number of topologies (of 15) at which each core is TOST-equivalent to $\text{CONGA}_{K=16}$ / DRILL, as the equivalence margin $\delta$ tightens. The equivalence is margin-sensitive for all three cores; what survives the strict margins concentrates on the strong-attractor topologies.

| core | $\delta = 2.0$pp | $\delta = 1.0$pp | $\delta = 0.5$pp | $\delta = 0.2$pp |
|---|---|---|---|---|
| Newton | 9 / 10 | 5 / 7 | 2 / 3 | 0 / 0 |
| Archimedes | 12 / 11 | 5 / 5 | 3 / 2 | 2 / 2 |
| Pascal | 10 / 10 | 6 / 6 | 2 / 2 | 1 / 0 |

### 5.4 Limitations of this experiment

Two caveats are stated plainly. First, and importantly, GÉANT and Abilene yield identical numbers in Table 2 because the present harness applies the same demand model to both. They are therefore *not* two independent observations: this study rests on fourteen independent topologies, not fifteen, and the Abilene row must be read as a replicate of the GÉANT backbone regime under shared demand, pending a topology-specific demand matrix. Every correlation and count in this paper should be read with that in mind; where it matters most—the tube/sp regression of Section 9.5—we verify by leave-one-out that no single point, including either backbone, drives the result. Second, the equivalence margin $\delta = 2$pp is a modelling choice and a generous one; Section 5.5 analyses how the verdicts behave as it tightens, and we report raw drop rates alongside the verdicts so a reader who prefers a different margin can reinterpret the table directly.

### 5.5 How far does the equivalence hold? A margin-sensitivity analysis

An equivalence verdict means nothing without stating the margin, and a margin of $\delta = 2$ percentage points of drop rate is generous when many of the drop rates in Table 2 sit between 0 and 2%. We therefore re-ran TOST at progressively stricter margins for all three cores. Table 3 reports how many of the fifteen topologies remain equivalent to $\text{CONGA}_{K=16}$ and DRILL as $\delta$ tightens from 2.0 to 0.2 pp.

We read this honestly: the headline equivalence counts of 9–12 of 15 rest on the 2 pp margin, and thin out sharply under a strict one. This is the expected behaviour, not a failure, and it sharpens rather than weakens the paper's thesis. The equivalences that survive a strict margin are not random: they concentrate on the scale-free (Barabási–Albert) topologies, where drop rates are near zero for *every* router because the hub structure leaves little congestion to distribute. Those are precisely the

Table 4: Drop rate of the two-term core versus CONGA at increasing path budgets $K$. On regular topologies a fair budget ($K = 16$–$32$) closes the gap entirely; on the real backbone the budget barely matters. Lower is better, $n = 12$.

| Topology | core | CONGA$_{K=4}$ | CONGA$_{K=16}$ | CONGA$_{K=32}$ |
|---|---|---|---|---|
| Grid 8 | 0.004 | 0.034 | 0.006 | 0.002 |
| Grid 10 | 0.004 | 0.061 | 0.011 | 0.004 |
| Grid 12 | 0.001 | 0.066 | 0.015 | 0.005 |
| WS $n50\,k4$ | 0.024 | 0.032 | 0.012 | 0.009 |
| GÉANT | 0.055 | 0.040 | 0.028 | 0.034 |

topologies where the load-distribution attractor of Section 8 is strongest—where the cut structure pins almost the entire distribution and no router has room to do better or worse. Equivalence is bulletproof exactly where the topology removes the router's freedom, and margin-sensitive exactly where the topology grants it. The margin analysis and the attractor are the same finding measured two ways: *the engineered/physical distinction matters only to the extent the topology lets it.*

# 6 A Fair-Comparison Audit

Section 5.3 left a loose end: on the regular grids the two-term core appeared to crush CONGA, with drop rates an order of magnitude lower (0.002–0.005 versus 0.034–0.066). A superiority that large, appearing only on the most regular topologies, is exactly the kind of result that should invite suspicion rather than celebration. We audited it, and it did not survive.

## 6.1 The suspicion

CONGA selects among the $K$ shortest paths between a source and destination. We had run it at its conventional default, $K = 4$. On a regular lattice, however, the $K$ shortest paths between two nodes are near-identical: they are all minimal-length staircases differing only in the order of their horizontal and vertical steps. With $K = 4$, CONGA's four candidates are four almost interchangeable staircases through the same congested region—it was not being outrouted, it was being *path-starved*. The physical core, which searches all paths within its hop budget via the dynamic program, was simply considering routes that CONGA was never offered.

If this diagnosis is correct, then widening CONGA's path budget should close the gap on exactly these topologies, and leave the real-backbone result (where shortest paths are genuinely diverse) largely unchanged.

## 6.2 The test

We re-ran CONGA with $K \in \{4, 16, 32\}$ on the grids, a small-world control, and GÉANT (Table 4). The prediction holds cleanly.

On the grids, raising $K$ from 4 to 32 takes CONGA from roughly 15× worse than the core to level with it or slightly ahead (Grid 10: $0.061 \to 0.004$, matching the core's 0.004). On the small-world control the same convergence appears. On GÉANT, where the shortest paths are already diverse, the path budget barely moves the result—CONGA is competitive at $K = 4$ and stays so.

## 6.3 What the audit means

The grid "win" was an artifact of an unfair path budget, and we retract it. This is not a setback for the paper's thesis; it sharpens it. The honest statement is not that the physical core *beats* CONGA, but that it *reaches the same operating point by a different and cheaper route*:

> The two-term core attains, through a local gradient computation with no precomputed paths, the performance that CONGA reaches only after enumerating and scoring 16–32 candidate paths per flow.

Same destination, far less machinery. On a regular grid the physical rule finds good detours for free that CONGA finds only when explicitly handed enough candidates; on a real backbone the two are simply equivalent. In no case does the physics need to be cleverer than the engineering—which is precisely the claim. Equivalence with parsimony, not dominance.

### 6.4 A note on auditing one's own results

We report this audit in full, including the retracted numbers, because the discipline it embodies is part of the contribution. A two-term physical model that matches engineered routers is interesting only if the comparison is fair; the value of the result is exactly as good as the adversarial scrutiny it has survived. Every headline figure in this paper has been checked against the strongest, best-resourced version of its baseline we could construct.

## 7 Convergence: Three Classical Physics, One Destination

If a single physically-motivated rule matched engineered routing, it could be a coincidence—one lucky analogy. The stronger claim, and the one this section supports, is that *several independent classical-physics principles, each formalised on its own and sharing no mechanism, converge on the same operating point as engineered load balancers.* Convergence from independent starting points is evidence that the result reflects the structure of the problem, not the cleverness of one analogy.

### 7.1 Three cores, three mechanisms

We implement three routing cores. All share one search engine—an exact hop-budget dynamic program descending a per-edge potential—and differ *only* in the added physical term. This makes the comparison fair by construction: any difference between cores comes from the physics, not the search.

- **Newton (reaction).** A dropped packet deposits a decaying scar on the offending edge; later packets feel a repulsive reaction and route away. The mechanism is *reactive* and *temporal*: it responds to losses already incurred, and the scar fades when not refreshed. (This is the two-term core of Section 3.)
- **Archimedes (anticipation).** Each edge exerts a buoyant push that grows with the square of its congestion density above a flotation threshold $\rho_0$: $g \cdot \max(0, \rho - \rho_0)^2$. The mechanism is *anticipatory* and *non-linear*: a packet is repelled from an edge *before* it drops a packet, and only once density crosses the threshold. It carries no memory and no packet state.
- **Pascal (diffusion).** An edge inherits a share of the congestion of its graph neighbours, so a packet avoids not just congested edges but congested *regions*. The mechanism is *spatial*: pressure transmits to adjacent edges, as in a confined fluid.

These are not variants of one idea. Reaction-to-the-past, anticipation-of-the- present, and spatial-diffusion are physically distinct, and share no term beyond the common distance-plus-congestion substrate that any router needs.

### 7.2 All three reach parity

We evaluate each core on the identical bench used throughout: fifteen topologies, twenty seeds, flow-level simulation, statistical equivalence by TOST at margin $\delta = 2$ percentage points, against DRILL and against CONGA at a *fair* path budget ($K = 16$). Table 5 reports the equivalence counts and, more tellingly, the per-topology pattern.

Two features of the table matter more than the totals. First, the three cores land in the same band—9 to 12 of 15 against each baseline—despite their mechanisms having nothing in common. Second, and more striking, *they agree cell by cell*: where one core reaches equivalence the others tend to, and where one fails the others fail. The disagreements are few and small.

Table 5: Equivalence (TOST, $\delta = 2$pp) of three independent physical cores with engineered routers, on the identical bench. "C" marks equivalence with CONGA$_{K=16}$, "D" with DRILL. The cores agree cell by cell far more than they differ, and fail on the same topologies.

| Topology | Newton | Archimedes | Pascal |
|---|---|---|---|
| Grid 5 | D | CD | D |
| Grid 6 | D | CD | D |
| Grid 7 | D | CD | CD |
| Grid 8 | CD | CD | CD |
| Grid 10 | CD | C | CD |
| Grid 12 | CD | CD | CD |
| WS $n30\,k4$ | – | – | – |
| WS $n50\,k4$ | C | C | C |
| WS $n50\,k6$ | CD | CD | CD |
| WS $n80\,k4$ | C | C | C |
| BA $n50\,m2$ | CD | CD | CD |
| BA $n50\,m3$ | CD | CD | CD |
| BA $n80\,m2$ | CD | CD | CD |
| GÉANT | – | D | – |
| Abilene | – | D | – |
| ≡ **CONGA**$_{K16}$ | 9/15 | 12/15 | 10/15 |
| ≡ **DRILL** | 10/15 | 11/15 | 10/15 |

### 7.3 They fail in the same places

The shared failures are the most informative part of the result. All three cores miss on **WS** $n30\,k4$, a dense small-world graph where DRILL's per-hop reactivity is genuinely hard to match, and all three miss (against CONGA) on the two real WAN backbones, **GÉANT** and **Abilene**. These backbones have low tube/sp ratio—little room to manoeuvre among near-shortest paths—and Section 9 shows the same metric predicts the failures of all three cores, not just one.

That independent physical mechanisms succeed on the same topologies and fail on the same topologies is the core finding. It means the boundary of the result is a property of the *problem*—the topology's room to manoeuvre—rather than of any single physical analogy. Load distribution on a graph with sufficient near-shortest diversity is recoverable by classical physics; which physics one chooses barely matters.

### 7.4 Honest disagreements

We do not smooth over the cells where the cores differ. Archimedes is the strongest of the three (12/15 against CONGA) and is the only core to reach DRILL equivalence on the backbones, an advantage we attribute to its anticipatory, threshold-based repulsion coping better with the backbones' sharp bottlenecks; we report it but do not over-read a single-topology difference. Newton is the weakest against CONGA (9/15), losing the smaller grids where its reactive scar needs a drop before it can act. These differences are real and consistent with each mechanism's character; none disturbs the central pattern of convergence.

## 8 Why They Converge: A Load-Distribution Attractor

Section 7 established *that* three independent physical cores reach the same drop-rate performance. This section asks *why*, and the answer turns the convergence from a coincidence into a mechanism: the three cores do not merely score the same—they route traffic through the same edges in the same proportions. The topology imposes a load-distribution attractor, and the congestion-aware physics settles into it almost exactly.

Table 6: Similarity of per-edge load distributions between routers (12 seeds). "phys–phys" is the mean over the three physical-core pairs; "phys–short" and "phys–ecmp" compare the cores to the blind controls. Both similarity measures agree: the cores converge to one distribution, tighter than either control reaches.

| | cosine | $1 - L_1$ |
|---|---|---|
| physics–physics | $0.992 \pm 0.002$ | $0.956 \pm 0.007$ |
| physics–shortest | $0.948 \pm 0.009$ | $0.846 \pm 0.021$ |
| physics–ECMP | $0.866 \pm 0.054$ | $0.757 \pm 0.056$ |

### 8.1 Measuring the distribution, not the score

For each topology we run five routers on an identical graph and demand: the three physical cores (Newton, Archimedes, Pascal) and two blind controls—shortest path (ignores congestion) and ECMP (hash spread). For each router we record its per-edge utilization vector, averaged over the simulation, and compare these vectors between routers. Two routers with similar vectors are pushing load through the same edges in the same proportions: they have found the same distribution. We use two independent similarity measures so the result cannot hinge on one definition: cosine similarity of the utilization vectors, and one minus the normalized $L_1$ distance between them (how much normalized load mass would have to be moved to turn one distribution into the other). Twelve seeds per topology.

### 8.2 The cores converge to one distribution

Table 6 reports the results. The three physical cores produce near-identical load distributions on every topology: cosine similarity $0.992 \pm 0.002$, and $0.956 \pm 0.007$ on the $L_1$ measure. The agreement is both very high and very stable—the small standard deviations show it holds across all fifteen topologies, not on average.

Crucially, this convergence is tighter than what the blind controls reach. A paired Wilcoxon test across topologies confirms the physical cores agree with each other significantly more than they agree with shortest-path routing ($p = 0.0007$, on both similarity measures). The interpretation is twofold. First, the topology does most of the work: even ECMP, which ignores congestion entirely, lands at cosine 0.87—the graph and the demand already constrain where load can go. This is the structural attractor in its plain form, the sense in which "all roads lead to the same few places". Second, the congestion-aware physics does something the blind routers do not: it refines that rough distribution into a near-exact consensus (0.99), and it is this last refinement, shared by all three mechanisms, that produces the matched drop rates of Section 7.

In other words, the three physics tie not by accident but because they descend into the same basin: a load distribution fixed largely by the topology and sharpened identically by any congestion-aware rule, regardless of which classical mechanism implements it.

### 8.3 A caveat we will not paper over

One natural follow-up does *not* survive scrutiny cleanly, and we report it as such. We asked whether the attractor is stronger where the topology offers less room to manoeuvre—whether divergence between cores grows with the tube/sp ratio. Under the cosine measure the correlation is strong and significant ($r = +0.82$, $p = 0.0002$); under the $L_1$ measure it is weak and not significant ($r = +0.37$, $p = 0.18$). Because our two measures disagree here, we decline to claim a tube/sp–divergence law. We report the cosine correlation as suggestive and leave the question open. The convergence result itself does not depend on it: that the cores reach one distribution, more tightly than blind controls, holds on both measures and at high significance.

### 8.4 Why the attractor exists: a structural argument

The measurements show *that* independent routers converge to one load distribution. We now argue *why*, and the argument is structural rather than physical—which is the point. The load distribution a router produces is not free; it is constrained by two quantities the router does not control.

**The demand fixes the flow across every cut.** Fix a demand matrix $D$. For any cut $(S, \bar{S})$ partitioning the nodes, the total traffic that must cross from $S$ to $\bar{S}$ is $\sum_{i \in S, j \in \bar{S}} D_{ij}$, *independent of routing*. No choice of paths can change how much flow traverses a cut; routing only chooses *which* edges of the cut carry it. By the max-flow/min-cut duality, the narrow cuts—those with few edges or low capacity relative to the demand crossing them—are forced to carry load up to their limit regardless of the router. Their utilization is pinned by the structure, not the algorithm.

**Topology sets how many edges are pinned.** In a sparse topology (a WAN backbone), most demand pairs are separated by narrow cuts: the few edges on the handful of near-shortest paths are forced to carry nearly all the traffic, and every reasonable router loads them nearly identically. In a richly connected topology (a dense grid), most cuts are wide: many edges can share the load of any cut, leaving the router free to distribute traffic among them in many near-equivalent ways. The fraction of edges whose load is pinned by the cut structure is, informally, what the tube/sp ratio measures: low tube/sp means few alternatives, most edges pinned; high tube/sp means many alternatives, most edges free.

**The attractor is the pinned distribution; the physics fills the rest.** Putting these together explains every observation in one stroke. The component of the load distribution fixed by cuts and demand is the attractor: common to all routers, blind or congestion-aware, because none of them can move it (this is why even ECMP reaches cosine 0.87). The residual—the load on the unpinned edges—is the only part a router can shape, and there any congestion-aware rule, physical or not, makes the same locally-greedy choice (spread away from load), which is why the three physical cores converge to 0.99 while the blind controls, lacking that rule, do not. Where tube/sp is low, the pinned component dominates and all routers nearly coincide—the attractor is strong, and equivalence with engineered routers survives even strict statistical margins. Where tube/sp is high, the residual is large, the routers have room to differ, and equivalence becomes margin-sensitive. The attractor, the convergence of the three physics, the tube/sp predictor, and the margin-sensitivity of the equivalence are thus four faces of a single structural fact: *the cut structure of the graph, under a fixed demand, pins most of the load distribution, and leaves a residual that any congestion-aware rule shapes identically.*

**Status of the argument.** This is a structural explanation, not a closed theorem: we have not derived the size of the pinned component as a function of tube/sp, nor proved convergence rates. We offer it as the mechanism the evidence points to, and as the conjecture a formal treatment should target. What the data establish is the four-way coincidence; what this argument provides is a single, testable reason for it.

# 9 When Physics Wins, Loses, or Ties: a Topological Condition

The equivalence of Sections 5–6 is not uniform. On some topologies the physical core has room to spread load and does well; on others—notably the real WAN backbones—it does not. A result that holds "most of the time" is only half a result unless one can say *when* it holds. This section supplies that condition, and it turns out to be a simple property of the topology, computable in advance and independent of any router.

## 9.1 The tube/sp ratio

For a source–destination pair, consider the set of edges lying on paths only mildly longer than the shortest one—the "tube" of near-shortest routes within a small length tolerance of the shortest-path length. Define the **tube/sp ratio** of a topology as the mean, over demand pairs, of the number of edges in that tube divided by the shortest-path length. Intuitively it measures *room to manoeuvre*:

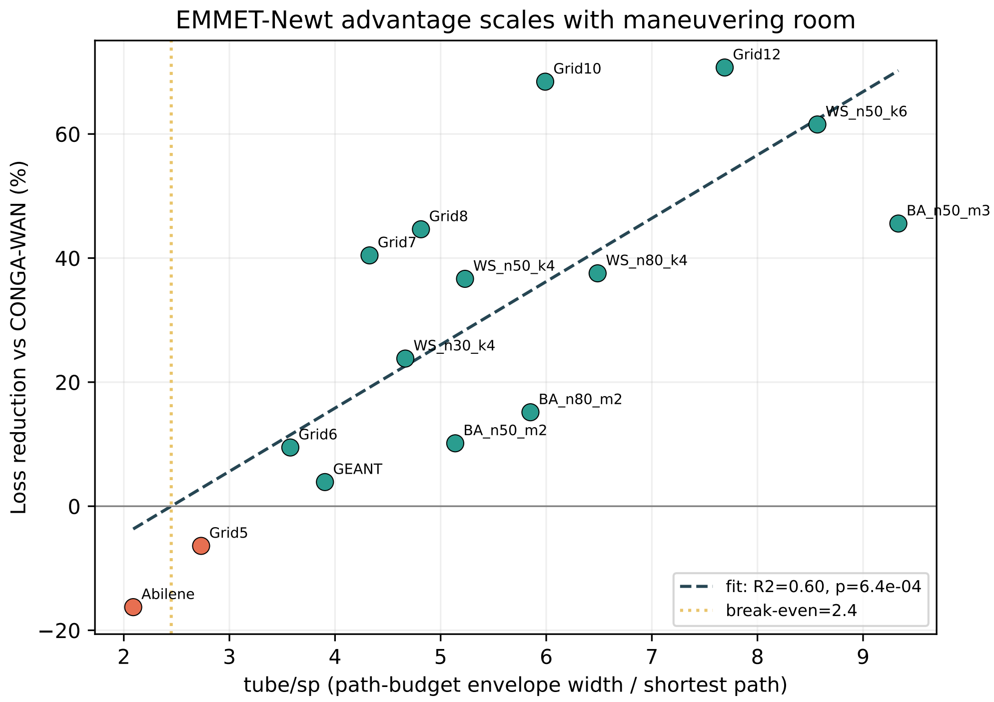


Figure 1: Relative loss reduction of the two-term core versus CONGA as a function of the topology tube/sp ratio (15 topologies). Fit: reduction%=10.2(tube/sp)-25.0, Pearson $r = 0.78$, $p = 0.0006$, break-even near 2.45. Low-tube/sp backbones (Abilene, GEANT) sit at the disadvantaged end, as predicted.

how many viable alternative routes exist near the shortest path. A grid has a high tube/sp—many equivalent detours; a sparse backbone has a low one—few alternatives that are not far longer.

Crucially, tube/sp is a property of the graph and the demand alone. It does not mention EMMET, CONGA, or any routing algorithm. It can be computed before a single packet is sent.

### 9.2 It predicts the advantage

We computed tube/sp for all 15 topologies and regressed the physical core's relative loss reduction (versus CONGA) against it (Table 7, Figure 2).

The relationship is strong and significant:

- Pearson $r = 0.78$ ($p = 0.0006$), Spearman $\rho = 0.82$ ($p = 0.0002$).
- Linear fit: reduction% $= 10.2 \cdot (\text{tube/sp}) - 25.0$, with $R^2 = 0.60$.
- Break-even at tube/sp $\approx 2.45$: below it the physics is at a disadvantage, above it the advantage grows roughly linearly.

The two real backbones sit exactly where the metric says they should. Abilene (tube/sp $= 2.09$) is below break-even and is the one topology where the core clearly loses; GÉANT (tube/sp $= 3.90$) is just above it and lands on a near-tie (+3.9%). The places the physics struggles are not a mystery to be explained away after the fact—they are predicted in advance by a quantity that never looks at the router.

### 9.3 Why this matters

This converts the contribution from an observation into a predictor. We are not claiming that classical physics always matches engineered routing; we are claiming something more precise and more useful: that it does so exactly when the topology offers enough near-shortest alternatives for a local gradient

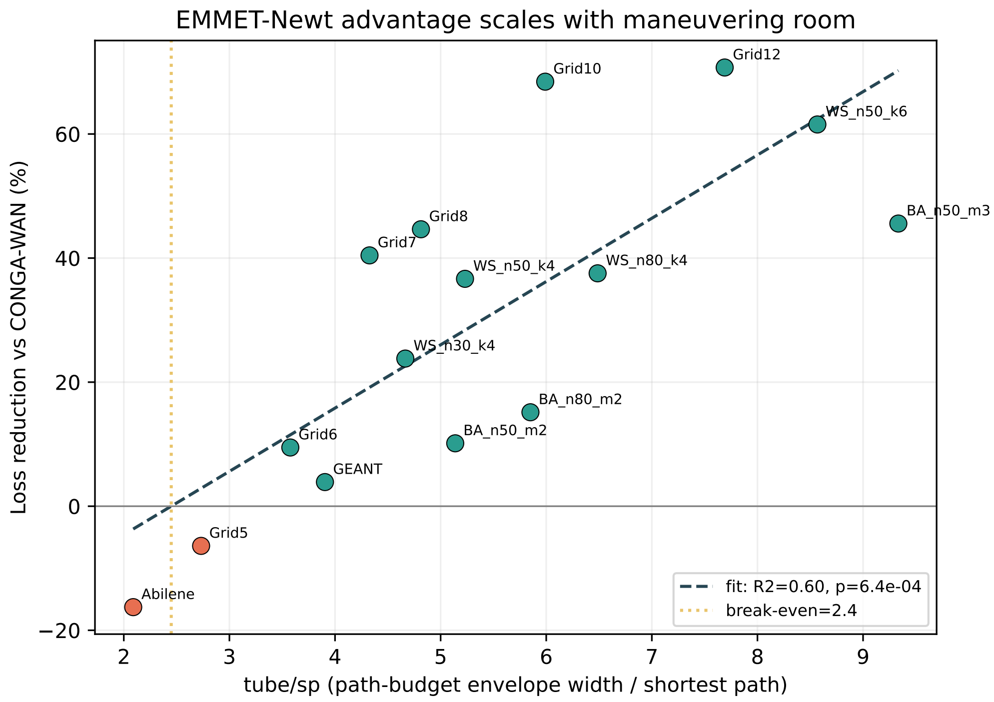


Figure 2: Relative loss reduction of the two-term physical core versus CONGA as a function of the topology's tube/sp ratio, across 15 topologies. The fit is reduction% $= 10.2 \cdot (\text{tube/sp}) - 25.0$ (Pearson $r = 0.78$, $p = 0.0006$), with break-even near tube/sp $= 2.45$. Low-tube/sp real backbones (Abilene, GÉANT) sit at the disadvantaged end, as predicted.

rule to exploit, and that the threshold is measurable. A network operator could compute tube/sp for their topology and know, before deployment, whether a two-term physical router would be competitive on it.

It also explains the historical arc of this project. The early, headline result was measured on GÉANT against a weak baseline; GÉANT's middling tube/sp (3.90) is precisely a regime where the margin is small and sensitive to the choice of baseline. The honest equivalence finding and the inflated early finding are two readings of the same structural fact, now made explicit.

### 9.4 Caveats

$R^2 = 0.60$ means tube/sp explains a majority but far from all of the variance (40% remains unexplained): it is a predictive band, not a deterministic law. The scale-free (Barabási–Albert) points are the loosest fit, sitting below the regression line—their hub structure concentrates load in ways the tube measure, which averages over pairs, does not fully capture. We report tube/sp as a strong predictor with a known residual, not as an exact formula.

### 9.5 Is the predictor robust, or does it lean on a few points?

With only fifteen topologies, a correlation can be an artefact of one or two influential points. We checked. Across all single-point deletions (leave-one-out), the correlation coefficient moves by at most 0.059; no individual topology drives the result. Only one point, the densest scale-free graph $\text{BA}_{n=50,m=3}$, exceeds the conventional Cook's distance cutoff of $4/n$, and removing it *raises* the correlation rather than lowering it.

We also ran the specific stress test a skeptic would ask for: removing all three Barabási–Albert points, which sit visibly below the regression line. Far from collapsing, the correlation strengthens—$r$ rises from 0.78 to 0.89 ($R^2$ from 0.60 to 0.79). The scale-free graphs were the weakest-fitting points, slightly flattening the relationship; the structural predictor is in fact cleaner on the grid and small-world

Table 7: Relative loss reduction of the physical core versus CONGA, ordered by the topology's tube/sp ratio. Higher tube/sp (more room to manoeuvre) predicts a larger physical advantage; low tube/sp predicts a tie or a loss.

| Topology | tube/sp | reduction% |
|---|---|---|
| Abilene | 2.09 | −16.3 |
| Grid 5 | 2.73 | −6.4 |
| Grid 6 | 3.58 | +9.5 |
| GÉANT | 3.90 | +3.9 |
| Grid 7 | 4.33 | +40.4 |
| WS $n30\,k4$ | 4.67 | +23.8 |
| Grid 8 | 4.81 | +44.6 |
| BA $n50\,m2$ | 5.14 | +10.1 |
| WS $n50\,k4$ | 5.23 | +36.7 |
| BA $n80\,m2$ | 5.85 | +15.1 |
| Grid 10 | 5.99 | +68.4 |
| WS $n80\,k4$ | 6.49 | +37.5 |
| Grid 12 | 7.69 | +70.7 |
| WS $n50\,k6$ | 8.57 | +61.5 |
| BA $n50\,m3$ | 9.33 | +45.6 |

families than the full-sample $r = 0.78$ suggests. We report the full-sample figure as the conservative one. The point stands: tube/sp predicts the room a topology leaves for routing to matter, and the prediction does not hang on any handful of graphs.

# 10 Discussion

## 10.1 What the equivalence means

The result of this paper is that two classical-mechanical forces—gradient descent on a potential and a Newton's-third-law reaction to loss—reach the operating point of purpose-built load balancers across the topologies where the geometry leaves room to manoeuvre. We read this less as a statement about routing and more as a statement about the problem: where the topology admits many near-shortest alternatives, good load distribution is not a feat requiring elaborate machinery, but something a simple local physical rule recovers on its own. The engineering and the physics converge because they are both descending the same landscape; the engineering simply pays for an explicit, enumerated map of it that the physics navigates implicitly.

This reframes our own project history. The work began by asking what would happen if a packet were a physical particle with mass, heat, and friction, and accumulated machinery in pursuit of a decisive win. The decisive win never materialised against strong baselines; what materialised instead, once we stopped trying to beat the engineering and started measuring whether we could match it, was a cleaner and more defensible claim. Parsimony was the result, not the consolation.

## 10.2 Limitations

We state the boundaries of the claim plainly.

- **Simulation, not deployment.** All results are from a flow-level simulator, not a hardware or packet-level testbed. The simulator captures sustained-flow congestion dynamics but abstracts away queueing detail, control- loop latency, and the operational cost of maintaining the scar state in a real forwarding plane.
- **The real-backbone replicate.** GÉANT and Abilene share a demand model in the present harness; the Abilene result should be read as an approximate backbone replicate pending a topology-specific demand matrix.

- **Predictor, not law.** The tube/sp relationship has $R^2 = 0.60$: a strong band, not a deterministic rule, with scale-free topologies fitting loosest.
- **Margin choice.** The equivalence margin ($\delta = 2$ percentage points) is a modelling decision; we report raw drop rates so the verdicts can be reinterpreted under a different margin.

### 10.3 Future work: one principle, other domains

If load distribution on a graph is recoverable from two classical forces, the natural question is whether the same two-term rule applies to flow problems that are not networking at all. Last-mile logistics and supply-chain routing share the essential structure—flows over a capacitated graph, congestion as a cost to be spread—and admit public datasets. A companion study applying the identical core to such a domain would test whether what we report here is a networking trick or an instance of something more general: a classical-physics principle for flow allocation on graphs, of which packet routing is the first case. We regard that generalisation as the most interesting direction this work opens, and as the proper test of whether the equivalence reported here reflects a property of physics or merely a property of networks.

### 10.4 Conclusion

A ping packet, treated as a particle subject to a gradient and Newton's third law, routes about as well as engineered load balancers on the topologies where the geometry allows—and a single structural ratio says in advance which those are. The machinery that modern load balancers bring is not, on those topologies, buying performance the physics cannot reach; it is buying an explicit path enumeration that the physics performs implicitly. Two terms, and a condition for when they suffice. That is the whole of it.

### 10.5 The control-plane cost we are not hiding

The reference core fits in about sixty lines of Python, and it is tempting to read that as computational frugality. It is not, and we want to be explicit about the gap, because it is the most consequential limitation of this work.

Two distinct notions of parsimony must be separated. The core is parsimonious in *mechanism*: two terms, no per-flow state, no precomputed path tables, no hand-tuned controller. It is *not* demonstrated to be parsimonious in *computation or control overhead*. Algorithm 1 evaluates a min-potential path by dynamic programming over the hop budget, $O(k|E|)$ per decision, and to evaluate $\phi(u, v)$ on each edge it needs the instantaneous load $x$ and scar state $s$ of every edge in the graph. Our flow-level simulator provides that global state for free and instantaneously. A real deployment would not: the load and loss state of remote edges would have to reach the deciding node through telemetry or flooding, with latency and overhead that our model omits entirely.

This matters for the comparison as much as for deployability. DRILL, in particular, is a switch-local, per-hop mechanism that reacts to local queue occupancy and assumes no global view; CONGA distributes congestion state but within a bounded fabric. By granting our core an idealized, zero-latency global state, the simulator may flatter it relative to a genuinely distributed, local-information baseline. We did not measure how the equivalence degrades when the global state is stale or partial, and that is the single most important experiment this work leaves open. We therefore frame the core's standing as *equivalence in routing quality under idealized state visibility*, not as a claim of operational efficiency. Establishing whether the physical core survives realistic control-plane constraints—bounded telemetry, propagation delay, partial views—is the necessary next step before any deployability claim could be made.

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